SIEMENS
Ingenuity for life

Acyclic communication between S7-1200 with V90 PTI via DS47

SINAMICS V90 PTI / V1.0 / Acyclic communication

https://support.industry.siemens.com/cs/ww/en/view/109758557

Siemens Industry Online Support

Legal information

Use of application examples

Application examples illustrate the solution of automation tasks through an interaction of several components in the form of text, graphics and/or software modules. The application examples are a free service by Siemens AG and/or a subsidiary of Siemens AG (“Siemens”). They are non-binding and make no claim to completeness or functionality regarding configuration and equipment. The application examples merely offer help with typical tasks; they do not constitute customer-specific solutions. You yourself are responsible for the proper and safe operation of the products in accordance with applicable regulations and must also check the function of the respective application example and customize it for your system.

Siemens grants you the non-exclusive, non-sublicensable and non-transferable right to have the application examples used by technically trained personnel. Any change to the application examples is your responsibility. Sharing the application examples with third parties or copying the application examples or excerpts thereof is permitted only in combination with your own products. The application examples are not required to undergo the customary tests and quality inspections of a chargeable product; they may have functional and performance defects as well as errors. It is your responsibility to use them in such a manner that any malfunctions that may occur do not result in property damage or injury to persons.

Disclaimer of liability

Siemens shall not assume any liability, for any legal reason whatsoever, including, without limitation, liability for the usability, availability, completeness and freedom from defects of the application examples as well as for related information, configuration and performance data and any damage caused thereby. This shall not apply in cases of mandatory liability, for example under the German Product Liability Act, or in cases of intent, gross negligence, or culpable loss of life, bodily injury or damage to health, non-compliance with a guarantee, fraudulent non-disclosure of a defect, or culpable breach of material contractual obligations. Claims for damages arising from a breach of material contractual obligations shall however be limited to the foreseeable damage typical of the type of agreement, unless liability arises from intent or gross negligence or is based on loss of life, bodily injury or damage to health. The foregoing provisions do not imply any change in the burden of proof to your detriment. You shall indemnify Siemens against existing or future claims of third parties in this connection except where Siemens is mandatorily liable.

By using the application examples you acknowledge that Siemens cannot be held liable for any damage beyond the liability provisions described.

Other information

Siemens reserves the right to make changes to the application examples at any time without notice. In case of discrepancies between the suggestions in the application examples and other Siemens publications such as catalogs, the content of the other documentation shall have precedence.

The Siemens terms of use (https://support.industry.siemens.com) shall also apply.

Security information

Siemens provides products and solutions with industrial security functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens’ products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the Internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial security measures that may be implemented, please visit https://www.siemens.com/industrialsecurity.

Siemens’ products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer’s exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Security RSS Feed at: https://www.siemens.com/industrialsecurity.

Table of contents

1 Task

Introduction

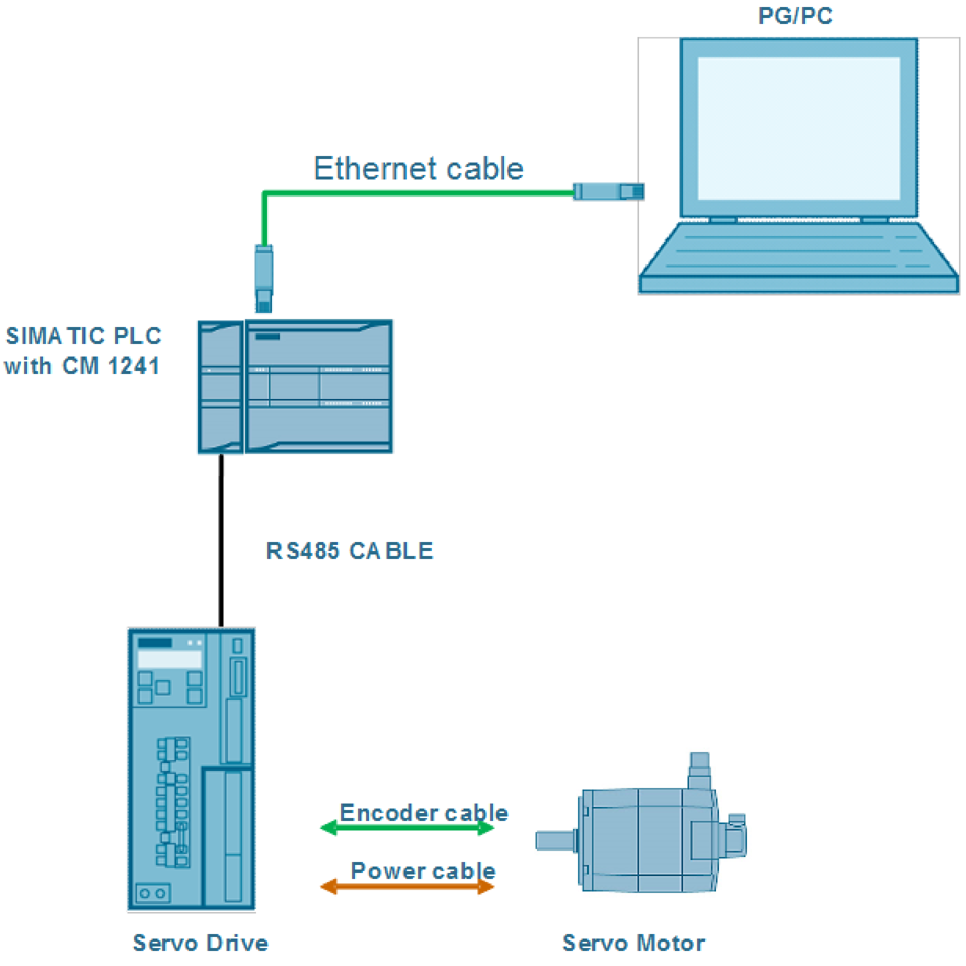
The SINAMICS V90 PTI servo drive supports the acyclic communication via data set 47. The SINAMICS V90 PTI drive can communicate with the PLC through an RS485 cable with the Modbus RTU communication protocol.

The S7-1200 PLC can read or write multiple parameters through acyclic communication using the communication module CM1241.

Overview of the automation task

The figure 1-1 below provides an overview of the automation task.

Figure 1-1

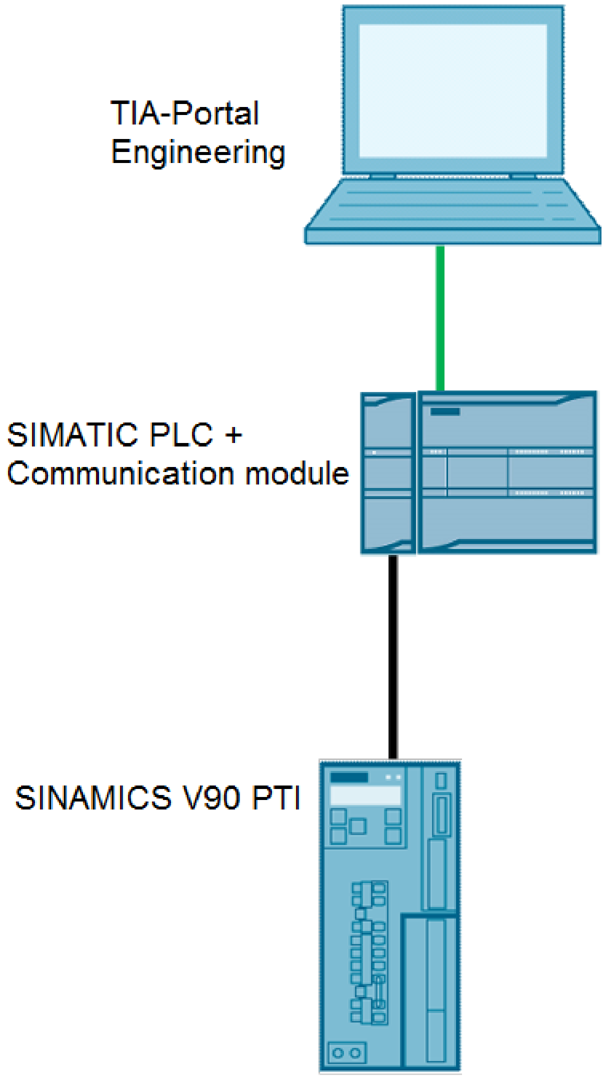


2 Solution

Schema Display

The following figure 2-1 displays the most important components of the solution:

Figure 2-1



Required knowledge

Basic knowledge about TIA Portal is assumed.

The configuration of S7-1200 PLC with the SINAMICS V90 PTI drive in TIA Portal is assumed.

The Modbus RTU communication knowledge and configuration of SINAMICS V90 PTI with S7-1200 PLC is assumed.

2.1 Hardware and Software Components

2.1.1 Validity

This application example is valid for

- TIA Portal V15
- S7-1200 CPU
- CM 1241 PtP
- SINAMICS V90 PTI FW V01.07.01
- SIMOTICS S-1FL6 HI motor

2.1.2 Used Components

The application was generated with the following components:

Hardware components

Table 2-1 hardware components

Component	No.	Article number	Note
SIMATIC S7-1200 CPU1217C DC/DC/DC	1	6ES7 217-1AG40-0XB0	V4.1
CM 1241	1	6ES7 241-1CH32-0XB0	
SINAMICS V90 PTI 400V	1	6SL3210-5FE10-4UA0	V1.7.1
SIMOTICS S-1FL6 Hi motor	1	1FL6042-1AF61-0LG1	400W

Standard software components

Table 2-2 software components

Component	No.	Article number	Note
TIA Portal	1		V15
SINAMICS V-ASSISTANT	1		V1.05.01

Sample files and projects

The table 2-3 includes all files and projects that are used in this example.

Table 2-3

Component	Note
109758557_V90 PTI Modbus Acyclic communication_DOC_en.pdf	Documentation
109758557_V90 PTI Modbus Acyclic communication_PROJ.zip	TIA Portal V15 Project

3 Acyclic data exchange

3.1 Description

The acyclic data exchange mode allows the following task:

- Exchange large “user data”.
- By using the PLC function blocks MB_MASTER, the user can exchange the data. The transferred data block structure should follow the DS47 data structure.

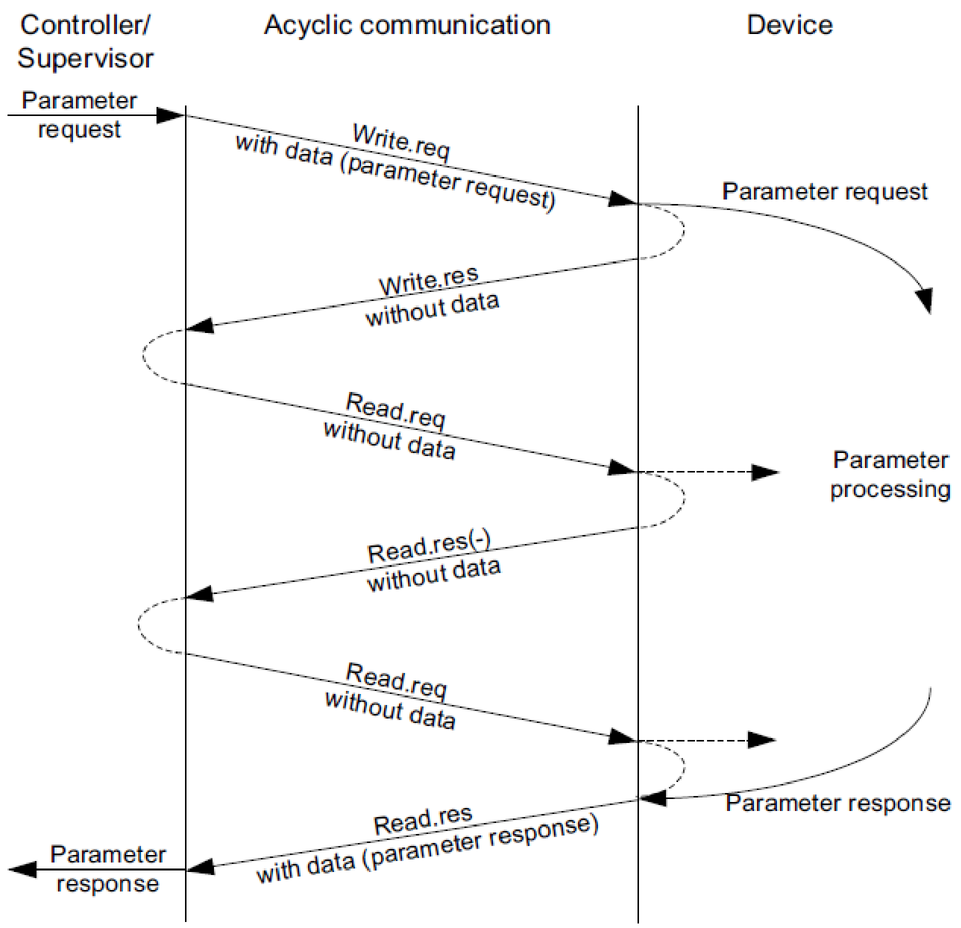
The quantity of the read or write parameter is not restricted. But the maximum data length is 240 byte for each read or write job.

Via FC16, up to 122 resisters can be written to the V90 PTI drive directly one after the other with one request. Register 40601 is for DS47 control, register 40602 is for DS47 header and register 40603 to 40722 is for DS47 data.

3.2 Diagram of acyclic data exchange

The figure 3-1 describes the principle of acyclic data exchange.

Figure 3-1



3.3 Read Parameter Data Structure

The table 3-1 below formats a request to read parameters.

Table 3-1 data structure of the task “read parameter”

Data block	**Byte n**	**Byte n+1**	**n**
Header	Refer 00 hex … FF hex	01 hex: read task	0
	02 hex	number of the parameters (m) 01 hex … 27 hex	2
The address of parameter 1	Properties 10 hex: parameter value	Number of index: 00 hex … EA hex (parameter has no index: 00 hex)	4
	Parameter number: 0001 hex … FFFF hex		6
	The first index number: 0000 hex … FFFF hex (parameter has no index: 0000 hex)		8
The address of parameter 2	…		…
…	…		…
The address of parameter m	…		…

Table 3-2 response data structure of the task “read parameter”

Data block	**Byte n**	**Byte n+1**	**n**
Header	The same as table 3-1	01 hex: read task is finished 81 hex: read task is not finished	0
	02 hex	The same as table 3-1	2
The value of parameter 1	Value format: 02 hex: Integer 8 03 hex: Integer 16 04 hex: Integer 32 05 hex: Unsigned 8 06 hex: Unsigned 16 07 hex: Unsigned 32 08 hex: Floating Point 0A hex: Octet String 0D hex: Time Difference 41 hex: Byte 42 hex: Word 43 hex: Double Word 44 hex: Error	Number of index values or number of error values for a negative response.	4
	The value of the first index. It will be the error value if the response is negative.		6
	…		…
The value of parameter 2	…		…
…	…		…
The value of parameter m	…		…

3.4 Write Parameter Data Structure

The table 3-3 below formats a request to write parameters.

Table 3-3 data structure of the task “write parameter”

Data block	Byte n	Byte n+1	n
Header	Refer 00 hex … FF hex	02 hex: write task	0
	02 hex	number of the parameters (m) 01 hex … 27 hex	2
The address of parameter 1	10 hex: parameter value	number of the parameter index: 00 hex … EA hex (the meaning of 00 hex and 01 hex is the same)	4
	Parameter number: 0001 hex … FFFF hex		6
	The first index number: 0000 hex … FFFF hex (parameter has no index: 0000 hex)		8
The address of parameter 2	…		…
…	…		…
The address of parameter m	…		…
The value of parameter 1	Value format: 02 hex: Integer 8 03 hex: Integer 16 04 hex: Integer 32 05 hex: Unsigned 8 06 hex: Unsigned 16 07 hex: Unsigned 32 08 hex: Floating Point 0A hex: Octet String 0D hex: Time Difference 41 hex: Byte 42 hex: Word 43 hex: Double Word	number of the parameter index: 00 hex … EA hex	
	The value of the first index		
	…		
The value of parameter 2	…		…
…	…		…
The value of parameter m	…		…

It the V90 PTI drive finishes the write task, then the response data format is shown in table 3-4.

Table 3-4 response data structure of the task "write parameter"

Data block	Byte n	Byte n+1	n
Message header	The same as table 3-3	02 hex	0
	02 hex	The same as table 3-3	2

If the V90 PTI drive doesn't finish the write task, then the response data format is shown in table 3-5.

Table 3-5 response data structure of the task "write parameter"

Data block	Byte n	Byte n+1	n
Message header	The same as table 3-3	82 hex	0
	02 hex	The same as table 3-3	2
The value of parameter 1	Value format: 40 hex: Zero, the write task is executed 44 hex: Error, the write task is not executed	number of error 00 hex or 02 hex	4
	Error value 1 only in error status [1]		6
	Error value 2 only in error status. The value is 0 or the first index number		8
The value of parameter 2	…		…
…	…		…
The value of parameter m	…		…

1) The error value will be described later.

[1] The error value will be described later.

3.5 Error code

The table 3-6 will describe the error code in read and write job.

Table 3-6 error code

Error code	Contents
00 hex	Illegal parameter number
01 hex	Parameter value can't be changed.
02 hex	Lower or upper value limit exceeded.
03 hex	Incorrect sub-index.
04 hex	No array.
05 hex	Incorrect data type.
06 hex	Setting not permitted, only resetting.
07 hex	Descriptive element can't be changed.
09 hex	Description data not available.
0B hex	No master control.
0F hex	Text array doesn't exist.
11 hex	Request can't be executed due to the operating state.
14 hex	Inadmissible value.
15 hex	Response too long.
16 hex	Illegal parameter address.
17 hex	Illegal format.
18 hex	Number of values not consistent.
19 hex	Drive object doesn't exist.
20 hex	Parameter text can't be changed.
21 hex	Service is not supported.
6B hex	A change request for a controller that has been enabled is not possible.
6C hex	Unknown unit.
77 hex	Change request is not possible during download.
81 hex	Change request is not possible during download.
82 hex	Accepting the master control is inhibited.
83 hex	Desired interconnection is not possible.
84 hex	Drive does not accept a change request.
85 hex	No access methods defined.
87 hex	Know-how protection active, access locked.
C8 hex	Change request below the currently valid limit.
C9 hex	Change request above the currently valid limit.
CC hex	Change request not permitted.

4 Programming and Operation of this application

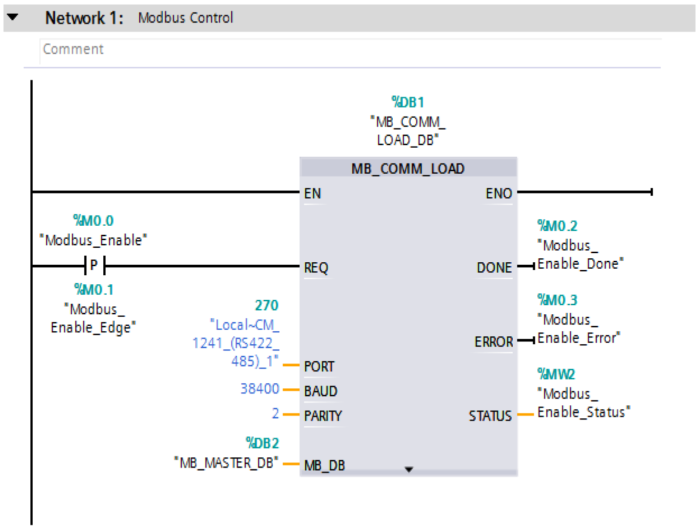
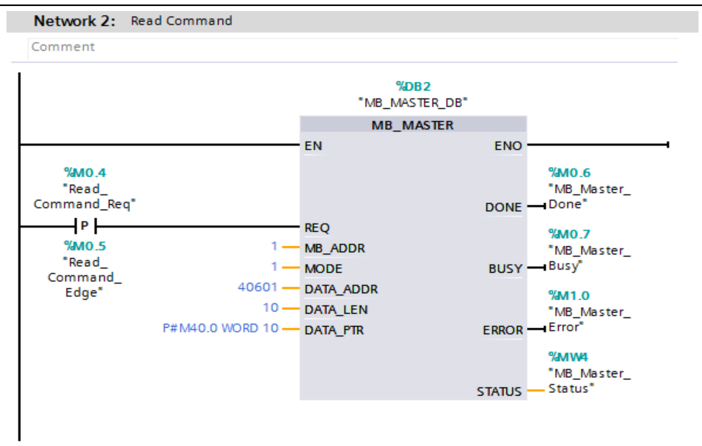
The details about how to do the configuration for the S7-1200 with V90 PTI is provided in the applications:

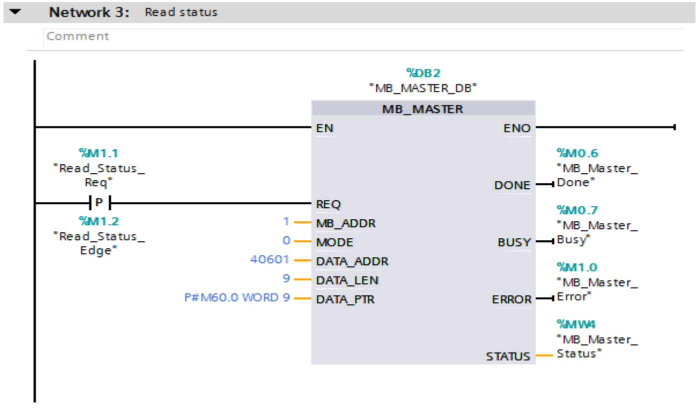
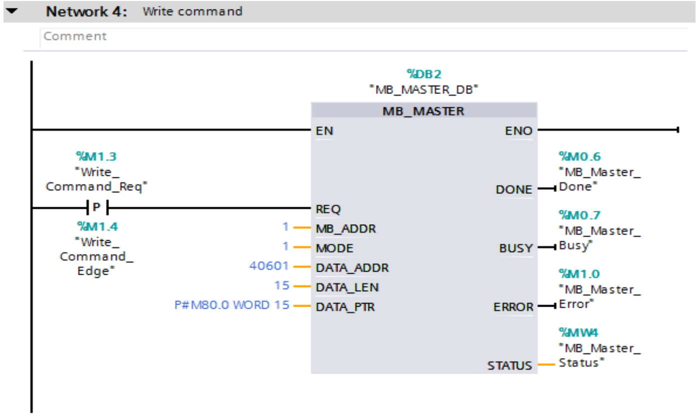
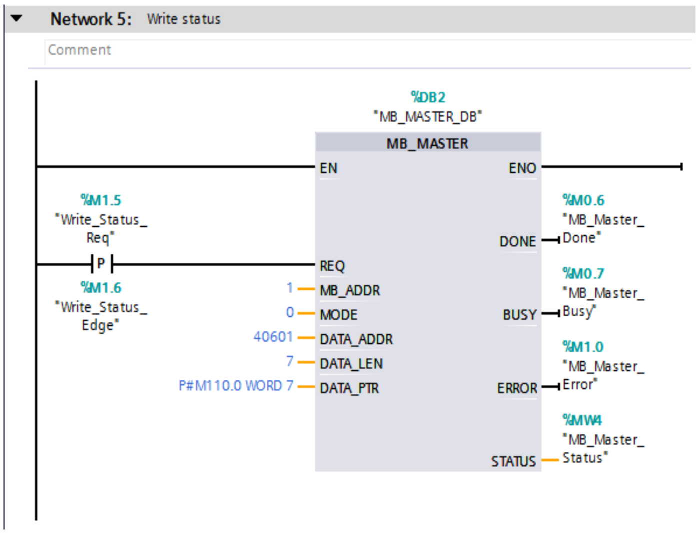
- You can download the application reference for Positioning (IPos) and Speed Control (S) with a V90via Modbus from the following link:
  https://support.industry.siemens.com/cs/ww/en/view/109480267

4.1 Configuration and programming

The configuration and programming is showed in table 4-1.

Table 4-1

Nr.	Action	Remarks
1		Initial the modbus communication.
2		Send the read command to the drive.

<table>
<tr><td>3</td><td>
</td><td>Receive the status of the read command.</td></tr>
<tr><td>4</td><td>
</td><td>Send the write command to the drive.</td></tr>
<tr><td>5</td><td>
</td><td>Receive the status of the send command.</td></tr>
</table>

4.2 Read multiple parameters

In this example the task is to read several parameters from the drive. The parameters are P1215 and P1120. Table 4-2 shows the operation sequence.

Table 4-2

Nr.	Action	Remarks
1	Set the M0.0=1 to initial the Modbus communication.	
2	According to the data structure shown in table 3-1, fulfill the value of MW40 to MW58. Shown in table 4-3.	
3	Set M0.1=1 to send the read command to the drive. After the execution finished, set M1.1=1 to receive the status of the read command.	
4	The received data shown in table 4-4.	

Table 4-3

	Byte n		Byte n+1		Address
Message header	DS47 control =1 activate request				MW40
	Function	2F hex	Request length	10 hex	MW42
	Request Ref	80 hex	Request ID	01 hex	MW44
	Drive ID	02 hex	Para. Quantity	02 hex	MW46
Parameter 1	Properties	10 hex	Index Quantity	01 hex	MW48
	Parameter number = 04BF hex				MW50
	The first index number = 0000 hex				MW52
Parameter 2	Properties	10 hex	Index Quantity	01 hex	MW54
	Parameter number =0460 hex				MW56
	The first index number = 0000 hex				MW58

Table 4-4

	Byte n		Byte n+1		Address
Message header	DS47 control =2 request was executed				MW60
	Function	2F hex	Request length	0E hex	MW62
	Request Ref	80 hex	Request ID	01 hex	MW64
	Drive ID	02 hex	Para. Quantity	02 hex	MW66
Parameter 1	Value format	03 hex	Index Quantity	01 hex	MW68
	Parameter value = 0				MW70
Parameter 2	Value format	08 hex	Index Quantity	01 hex	MW72
	Parameter value = 2.0				MD74

The result of the read task is: P1125=0, and P1121=2.0. It’s the same in the drive.

4.3 Write multiple parameters

In this example the task is to write several parameters from the drive. The parameters are P1121, P29300. Table 4-5 shows the operation sequence.

Table 4-5

Nr.	**Action**	**Remarks**
1	According to the data structure of table 3-3, fulfill MW80 to MW104. The format of the buffer shows in table 4-6.	
2	Set M1.3=1 to send the write command to the drive. After the execution finished, set M1.5=1 to receive the status of the write command.	
3	The received data shown in table 4-7.	

Table 4-6

	Byte n		Byte n+1		Address
Message header	DS47 control =1 activate request				MW80
	Function	2F hex	Request length	1A hex	MW82
	Request Ref	80 hex	Request ID	02 hex	MW84
	Drive ID	02 hex	Para. Quantity	02 hex	MW86
Parameter 1	Properties	10 hex	Index Quantity	01 hex	MW88
	Parameter number = 0461 hex				MW90
	The first index number = 0000 hex				MW92
Parameter 2	Properties	10 hex	Index Quantity	01 hex	MW94
	Parameter number = 71CA hex				MW96
	The first index number = 0000 hex				MW98
Value of parameter 1	Value format	08 hex	Index Quantity	01 hex	MW100
	Parameter Value = 11.28				MD102
Value of parameter 2	Value format	03 hex	Index Quantity	01 hex	MW106
	Parameter Value = 0002hex				MW108

Table 4-7

	Byte n		Byte n+1		Address
Message header	DS47 control =2 request was executed				MW110
	Function	2F hex	Request length	04 hex	MW112
	Request Ref	80 hex	Request ID	02 hex	MW114
	Drive ID	02 hex	Para. Quantity	02 hex	MW116

The result of the write task is: P1121=11.28, and P29300=2. Check in the drive, the parameter value is modified.

5 Related literature

Table 5-1

	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Download page of this entry https://support.industry.siemens.com/cs/ww/en/view/109758557
\3\	S7-1200 manual https://support.industry.siemens.com/cs/ww/en/view/91696622 • Chapter 12.1: Communication processor and Modbus TCP • Chapter 12.5 Modbus RTU and TCP • Chapter 12.5.3 Modbus RTU

6 Contact

Siemens Ltd., China

DF M3-BF GMC

No. 18 Siemens Road
Jiangning Development Zone

Nanjing, 211100
China
mailto: mc_gmc_mp_asia.cn@siemens.com

7 History

Table 7-1

Version	Date	Modifications
V1.0	06/2018	First version